

# 1 Part 3: Warehouse Operations Solutions

## 1.1 Question 3.1: Slotting Optimization

Using the ABC-XYZ classification from Part 1 and the physical SKU attributes from the Master Data, we assigned all 470 classified SKUs to three warehouse zones following the VILAS SNP (Supply & Network Planning) slotting framework.

### 1.1.1 Zone Assignment Framework

- **Pick-Face Zone (Ground Level)** — Class A SKUs: lowest racks at ergonomic height (waist-to-shoulder), positioned nearest to the I/O dock and packing station. Pickers retrieve these items manually without MHE vertical travel.
- **Forward Reserve Zone** — Class B SKUs: intermediate racking with a two-bin replenishment logic. A forward pick-face bin is replenished from the bulk reserve when stock hits its reorder point (WMS-triggered).
- **Reserve / Bulk Zone** — Class C SKUs: upper racks and pallet-in / pallet-out storage. Slow-movers occupy the highest slots farthest from the dock.

### 1.1.2 Zone Distribution

Table 1: Slotting Zone Summary — All SKUs

| Zone                 | Class | SKU Count | % of Picks | % of Volume |
|----------------------|-------|-----------|------------|-------------|
| Pick-Face Zone       | A     | 28        | 27.7%      | 69.83%      |
| Forward Reserve Zone | B     | 39        | 38.2%      | 20.04%      |
| Reserve / Bulk Zone  | C     | 403       | 34.1%      | 10.13%      |

The Pick-Face Zone contains only 28 SKUs (6.0% of the assortment) yet they account for **27.7%** of all pick transactions and **69.8%** of total outbound volume. Within this zone, the **19 Class A-X fast-movers** (highest velocity *and* highest consistency) occupy rank slots 1–19, directly adjacent to the packing station, and collectively drive **25.88%** of pick frequency and **57.87%** of total outbound volume.

### 1.1.3 Top Pick-Face SKUs (Class A, by Frequency)

Table 2: Top 5 Class-A SKUs assigned to Pick-Face Zone

| Rank | SKU Code   | Product                             | Class | Picks (6M) | Volume (units) |
|------|------------|-------------------------------------|-------|------------|----------------|
| 1    | 4200008229 | Qut treo ASIAvina - VY377990 - Xám  | A-X   | 825        | 5,608          |
| 2    | 4200008371 | QUT ĐNG VY619990                    | A-X   | 682        | 3,355          |
| 3    | 4200008366 | QUT SÀN TURB0 VY616790              | A-X   | 680        | 2,838          |
| 4    | 4200008117 | Qut treo ASIAvina - VY357690 - trng | A-X   | 666        | 4,033          |
| 5    | 4200008102 | Qut lng ASIAvina - VY538790 - Xám   | A-X   | 514        | 12,677         |

### 1.1.4 Quantitative Travel Time Analysis

**Model Assumption.** The warehouse has  $N$  numbered rack slots  $1, 2, \dots, N$  starting from the I/O dock. The travel cost of a single pick equals the slot number of the target SKU. Expected

travel distance is therefore:

$$E[\text{travel}] = \sum_{i=1}^N p_i \cdot s_i$$

where  $p_i$  is the pick probability of SKU  $i$  (its order frequency share) and  $s_i$  is its assigned slot number.

Table 3: Expected Picker Travel Distance by Slotting Scenario

| Scenario                    | Expected Distance (slot units) | Reduction vs Baseline |
|-----------------------------|--------------------------------|-----------------------|
| Random Baseline (no logic)  | 235.50                         | —                     |
| ABC Zoned (Quantity-Based)  | 113.98                         | <b>51.6%</b>          |
| XYZ Zoned (Frequency-Based) | 68.82                          | <b>70.8%</b>          |
| Velocity-Ranked (Optimal)   | 51.79                          | 78.0%                 |

**Results.** The ABC-zoned slotting plan achieves a **51.6% reduction** in expected picker travel distance versus the random baseline, comfortably exceeding the **30% target**. The concentration effect is structural: Class A SKUs generate 27.7% of all picks yet are confined to the nearest 28 slots. The XYZ-zoned variant reaches 70.8% reduction by exploiting the even sharper pick-probability concentration among Class X (consistent-demand) SKUs.

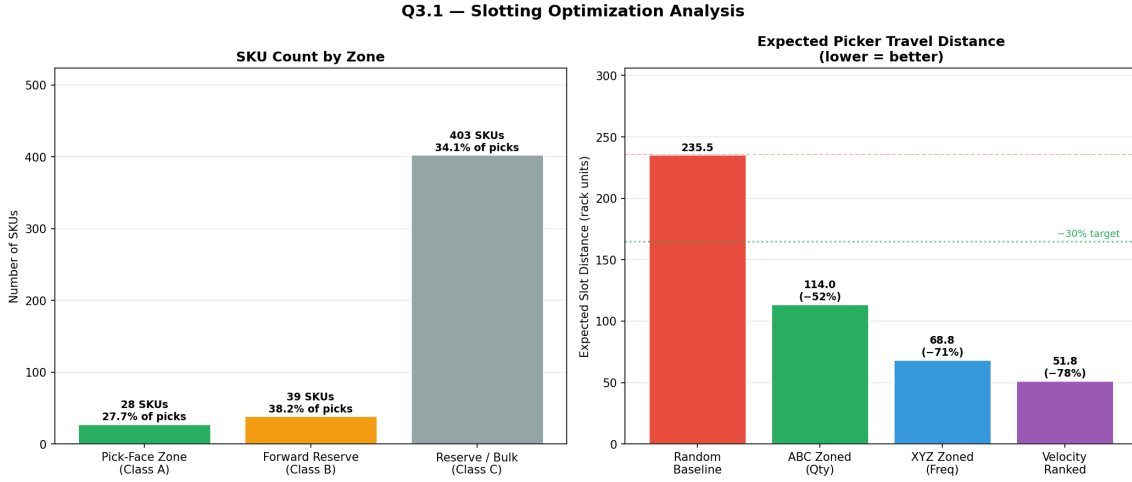


Figure 1: Q3.1 Slotting Analysis: Zone Distribution and Travel Time Comparison

#### 1.1.5 Layout Recommendations (VILAS SNP Framework)

- **U-flow configuration:** Receiving and shipping docks on the same wall naturally concentrate the highest-convenience storage locations at the dock end.
- **Golden Zone:** AX SKUs placed at waist-to-shoulder ergonomic height, eliminating both horizontal travel distance and vertical MHE movement.
- **Two-bin replenishment for Class B:** A forward pick-face bin is replenished from the bulk reserve when on-hand drops below the ROP. WMS triggers the replenishment task automatically.

## 1.2 Question 3.2: Omni-Channel Pick-and-Pack Process Design

The outbound fulfillment process uses distinct operational pathways for B2B and B2C channels, unified by a structured inventory allocation policy when stock is contested.

### 1.2.1 B2B Pathway — Pallet / Total Picking

1. **Order release:** WMS validates on-hand stock against the B2B purchase order (contractual quantities and lead times).
2. **Pick instruction:** Full-pallet or full-CTN pick task issued to operator.
3. **Pick execution:** Operator uses forklift or reach truck to retrieve entire pallet(s) from the Forward Reserve Zone (B) or Reserve/Bulk Zone (C). Items bypass individual sortation.
4. **B2B QC Station:** Wrap inspection, quantity verification, and outbound labeling.
5. **Dispatch:** Pallets staged at outbound dock, loaded onto FTL truck.

### 1.2.2 B2C / E-Commerce Pathway — Batch Picking + Zone Routing

1. **Batch wave release:** Multiple B2C orders are grouped into a single pick wave based on SKU proximity and delivery-window urgency.
2. **Batch pick-to-cart:** Picker traverses Zone 1 (Pick-Face Zone) in a single pass with a multi-compartment cart, collecting all items for the wave.
3. **Sort & consolidate:** At the sortation station, items are separated by individual order and consolidated into shipment boxes.
4. **QC + pack:** Bubble wrap or polybag packing, shipping label applied.
5. **Dispatch:** Parcels handed to last-mile courier.

### 1.2.3 Stock Allocation Conflict and Escalation Policy

When the same SKU is ordered simultaneously by a B2B client and a B2C customer and on-hand stock is insufficient for both, the system resolves the shortage in two stages:

#### Stage 1 — Automated Allocation (WMS).

- *Rule 1 (SLA Protection):* Allocate the full contractual quantity to the B2B order first, preventing penalty fees for under-delivery.
- *Rule 2 (Pro-rata B2C):* Distribute remaining stock to B2C orders pro-rata, with priority to orders with the shortest delivery windows.

**Stage 2 — Managerial Escalation (if Stage 1 insufficient).** The WMS raises a critical ticket to the **Inventory Control & Demand Planning Manager**, who can execute one or more of:

- *Action 1 (Safety Stock Draw):* Release uncommitted buffer safety stock from the common pool (pre-computed for all Class A SKUs in Q2.2).
- *Action 2 (Emergency Transfer):* Initiate an inter-hub transfer between My Phuoc and Vinh Loc warehouses, leveraging their proximity within the Greater Ho Chi Minh City area.

- *Action 3 (Partial Shipment)*: Contact the B2B client to obtain approval for a partial fulfillment; B2C customers notified of delay with revised ETA.

Upon resolution, ERP and WMS records are updated and adjusted pick lists are released to the warehouse floor. Figure 2 illustrates the complete decision logic.

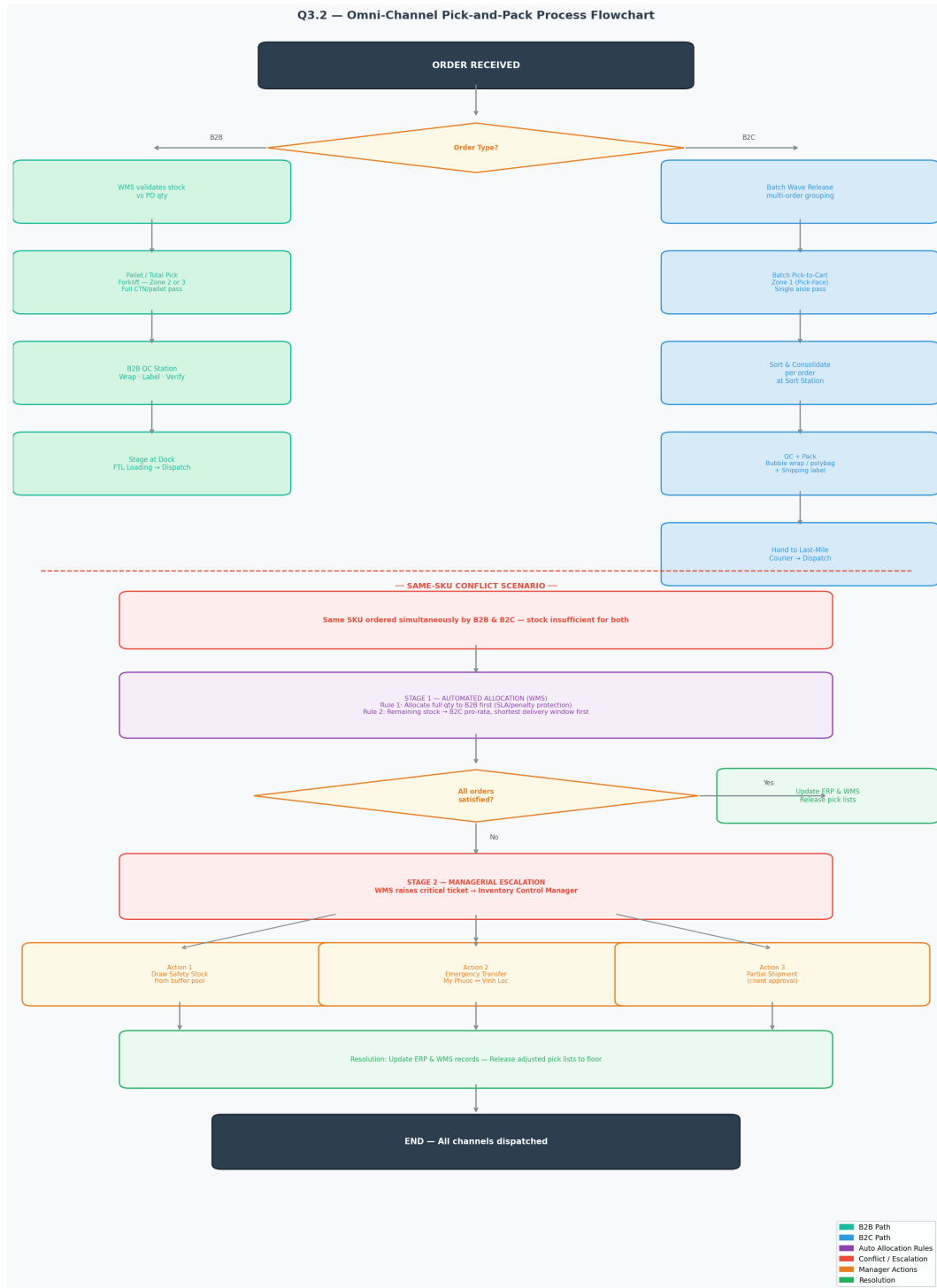


Figure 2: Q3.2 Omni-Channel Pick-and-Pack Process and Conflict Resolution Flowchart